

Characteristics and Outcomes of 237 Juxtapapillary Choroidal Melanomas Treated with Iodine-125 Plaque Brachytherapy

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Keywords

Uveal melanoma · Plaque brachytherapy · Radiation complications

Abstract

Introduction: Juxtapapillary uveal melanomas limit plaque brachytherapy treatment using the standard 2 mm margin and are often excluded from prospective studies thus limiting data on outcomes and complications. We retrospectively evaluated outcomes in this tumor population following primary iodine-125 plaque brachytherapy. **Methods:** We performed a retrospective review over 30 years of patients treated with iodine-125 plaque brachytherapy for juxtapapillary uveal melanoma at a single center. **Results:** Patients were white (97%), male (53.2%), 62 years old (median age, median follow-up of 4.1 years), with right eye involvement (54.4%). At 1, 5, and 10 years, local recurrence was observed in 2.2%, 10%, and 24.6%, enucleation in 0.05%, 11.3%, and 22.1%, metastasis in 1.3%, 6.7%, and 14.2%, and mortality in 1.8%, 14.9%, and 32.8%, respectively. Median visual acuity declined from LogMAR 0.1 to 2.0 at last visit. Radiation retinopathy and optic neuropathy were seen in 54.9% and 46%, respectively, of patients by a median of 742 days (2.03 years) and

1,011.5 days (2.77 years). **Conclusions:** Juxtapapillary melanomas demonstrated higher rates of vision loss and local recurrence following plaque brachytherapy compared with other tumor configurations. Enucleation, distant metastasis, and overall mortality were comparable to those reported for non-juxtapapillary melanomas.

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Introduction

Uveal melanoma is the most common primary intraocular malignancy in adults with an incidence of approximately 5.2 per million per year in the USA [1]. Juxtapapillary melanomas are typically defined as uveal melanomas that are adjacent to the optic nerve, specifically within 0- to 1-disc diameters of its margin based on previously published studies [2, 3]. Juxtapapillary tumors are difficult to treat due to a minimal or nonexistent posterior treatment margin which may increase the risk of local recurrence. The proximity of these tumors to the optic nerve and fovea also increases subsequent vision loss due to both radiation retinopathy and optic neuropathy. The majority of juxtapapillary tumors were excluded from the Collaborative Ocular Melanoma Study

(COMS) medium tumor trial [4, 5], limiting currently available prospective data.

Plaque brachytherapy remains the mainstay of treatment for medium tumors at most centers and is even prescribed for large tumors at some centers as well [6–8]. Traditional brachytherapy utilizes one of several radioisotopes including iodine-125 (I-125), ruthenium-106 (Ru-106), palladium-103 (Pd-103), or gold-79 (Au-79). At our institution, uveal melanomas are treated with a targeted apical dose of 85 Gy with a 2 mm margin around the base. Juxtapapillary tumors can be treated using a notched plaque, which slides around the optic nerve [2], allowing treatment to the optic nerve edge while increasing the risk of optic neuropathy. We present the treatment outcomes, rates of recurrence, globe salvage, metastasis, and all-cause mortality in a sequential series of juxtapapillary uveal melanoma patients treated with customized I-125 plaque brachytherapy at a single institution over 31 years.

Methods

A retrospective chart review was performed on 237 consecutive patients with juxtapapillary choroidal melanoma treated with primary I-125 plaque brachytherapy at a single institution from September 1989 to June 2020. Tumors included in the study had a margin between 0- and 1-disc diameters from the optic nerve or <1.5 mm from the optic nerve. All patients were prescribed an apical dose of 85 Gy delivered at 52.8 cGy/h over 7 days. The study was approved by the University of Tennessee Health Science Center Institutional Review Board and complies with the Declaration of Helsinki.

Plaque Design and Brachytherapy Treatment Protocol

I-125 seed model 2301 comprised uniform activity sources produced by Best Medical International, Inc. (Springfield, VA, USA) were utilized for all plaques. Radiation doses were computed following the brachytherapy dose formalism recommended by the American Association of Physicists in Medicine Task Group Report #43 [9]. Treatment dose was prescribed to the tumor apex or a minimum of 5 mm depth from the inner surface of the sclera, whichever was greater. The planned initial dose rate at the prescription point was 52.80 cGy/h with a planned implant duration of approximately 168 h (1 week), resulting in a cumulative prescription dose of 85 Gy.

Plaque design remained the same throughout the duration of the study. Plaques in standard sizes were

Table 1. Patient and tumor^{1,2} baseline characteristics

Age and follow-up	
Age at surgery, years	61 (14), 62
Follow-up, years	5.3 (4.6), 4.1
Sex	<i>n</i> (% of total)
Female	111 (46.8)
Male	126 (53.2)
Race	
White	215 (90.7%)
Non-White	6 (2.5%)
Unknown	16 (6.8%)
Laterality	
Left	108 (45.6%)
Right	129 (54.4%)
Smoker	
No	182 (76.8%)
Yes	44 (18.6%)
Unknown	11 (4.6%)
Tumor characteristics	
Apical height, mm	4.9 (3.2), 3.9
Max basal diameter, mm	12.5 (4.0), 12.1
Tumor volume, mm ³	263 (353), 136
AJCC tumor grade (T) ²	<i>n</i> (% of 235)
T1a	74 (31)
T2a	88 (37)
T2b	1 (0.4)
T3a	51 (22)
T4a	14 (6.0)
T4b	7 (3.0)
AJCC tumor stage	
I	74 (31%)
IIA	89 (38%)
IIB	52 (22%)
IIIA	13 (5.5%)
IIIB	7 (3.0%)

N = 237. Values are mean (standard deviation), median or number (percentage). ¹*N* = 235, 2 patients' tumor characteristics at baseline were unavailable. ²AJCC, American Joint Committee on Cancer, 8th edition.

selected to provide a minimum 2 mm margin around the tumor base. For tumors abutting the optic nerve or where a round plaque could not be placed without directly touching the optic nerve, plaques with a 6 mm wide notch were utilized allowing for placement around the optic nerve [2]. Radiation dose planning data were obtained from radiation oncology, and the dose to the optic nerve was extrapolated. Additional data on plaque size, configuration (notched vs. circular), and apical and scleral radiation dose were also recorded. Transpupillary

Table 2. Outcomes and complications

Outcomes and complications	Event, <i>n</i> (%)	Time to event, years
Recurrence	33 (14)	2.98 (2.26), 2.48
Enucleation	29 (12)	3.65 (3.50), 2.57
Metastasis	25 (11)	3.86 (2.70), 2.72
Mortality	52 (22)	5.1 (3.9), 4.2

Values are mean (standard deviation), median. Primary outcomes following iodine-125 plaque brachytherapy and time to primary outcomes. *N* = 237. SD, standard deviation.

thermotherapy was not performed in conjunction with brachytherapy as part of the primary treatment of the lesion.

Patient Outcome Data

Data on patient outcomes with regards to local tumor recurrence, extraocular extension, enucleation, metastasis, radiation optic neuropathy, radiation retinopathy, neovascular glaucoma, neovascularization of the iris, scleromalacia, cataract, vitreous hemorrhage, and retinal detachment were collected. Best corrected visual acuity (BCVA) was converted from Snellen to LogMAR values as previously described by Schulze-Bonsel and Bach [10]. All outcomes were censored at the time of last visit or death.

Long-Term Follow-Up Data

Follow-up interval was calculated from the date of plaque placement to time of last visit or death. All patients with a visit within the previous 1 year of study cessation were assumed to be alive unless otherwise notified. Patients who did not have a clinic visit within 1 year were searched using an online ancestry database (genealogybank.com) with public access to the Social Security Death Index (SSDI). Date of death was checked against the last appointment and date of plaque placement. Patients who were not identified as deceased within the SSDI were assumed to be alive and were censored at the date of last follow-up.

Statistical Analysis

The mean ± standard deviation and median were reported for continuous variables (e.g., age), while categorical variables (e.g., gender) were expressed as percent (n). A Student’s *t* test was used for continuous variables, and a Fisher’s exact test was used for categorical variables to determine the significant associations. The Kaplan-Meier method was utilized to calculate the 1-, 5-, and 10-

year survival estimates for overall survival, recurrence, enucleation, metastasis, and event-free survival (death or metastasis) for 25 years of follow-up. Both simple variate and multivariate Cox proportional hazard regression models were used to evaluate overall survival, event-free survival, and local recurrence, while controlling for demographic variables (age, gender, and race) and variables selected from the literature known to be associated with the respective outcome. Each model was then refined using the bi-directional stepwise method with Akaike information criterion for variable selection. The Cox proportional hazard assumption was tested using the Schoenfeld residue method. Local recurrence was used as a time-varying covariate in the multivariate analysis for event-free survival and overall survival analysis. Computations were performed using statistical software *R* (v 4.2.0; R Core Team [2022]) or *Stata* (*Stata/SE18.0*, College Station, TX, USA: StataCorp LLC.). Significance was set at *p* < 0.05 (two-sided).

Results

Patient and tumor baseline characteristics are summarized in Table 1. Two hundred thirty-seven patients with juxtapapillary uveal melanoma were included in the analysis. Median duration of follow-up was 49 months or 4.1 years (mean 5.3 years, range 0–25.1 years). Median tumor apical height was 3.9 mm with a median basal diameter of 12.1 mm.

Treatment outcomes and complications are summarized in Table 2. Tumor recurrence was noted in 33 patients. Kaplan-Meier estimate of local recurrence was 2.2% at 1 year, 10% at 5 years, and 24.6% at 10 years (Fig. 1a). Among those patients who developed local recurrence, the median time to recurrence was 2.48 years. Enucleation was performed in 29 patients. The Kaplan-Meier estimate of an enucleation event was 0.05% at

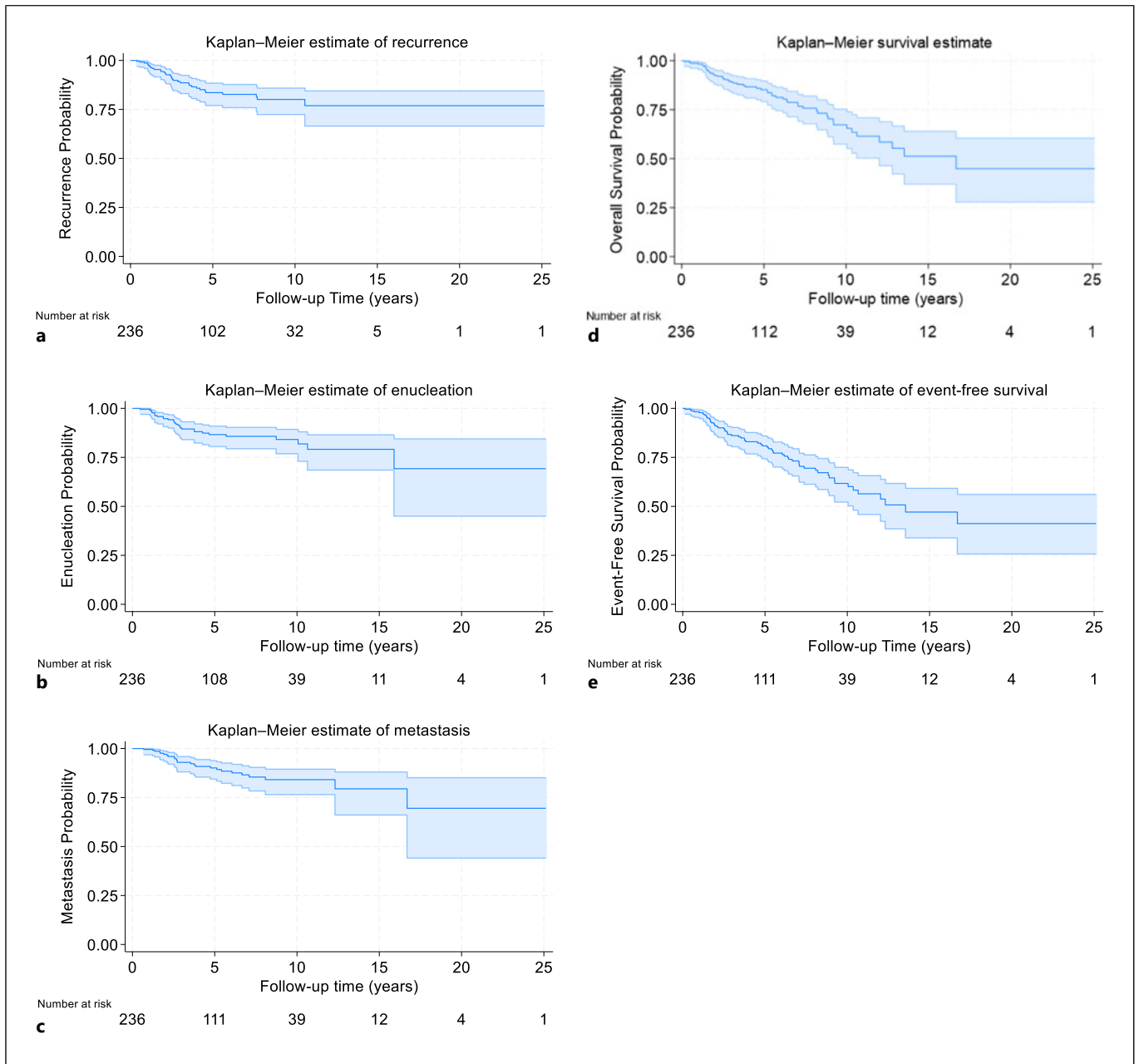


Fig. 1. Kaplan-Meier charts over 10 years of post-plaque brachytherapy for local recurrence rate (**a**), enucleation rate (**b**), distant metastasis rate (**c**), mortality rate (**d**), and event-free survival rate (metastasis or mortality) (**e**).

1 year, 11.3% at 5 years, and 22.1% at 10 years (Fig. 1b). The date of enucleation was missing for 4 patients. Reasons for enucleation included tumor recurrence ($n = 16$), blind painful eyes ($n = 10$), neovascular glaucoma ($n = 2$), and unknown ($n = 1$). Among patients who underwent enucleation, the median time to enucleation was 2.57 years.

Metastasis occurred in 25 (11%) patients. Kaplan-Meier estimate was 1.3% at 1 year, 6.7% at 5 years (6.7%), and 14.2% by 10 years (Fig. 1c). Time to metastasis was missing in 1 patient. Among those who had metastasis, median time to metastasis was 2.72 years. Overall rate of mortality was 22% for the entire cohort, with Kaplan-Meier estimates of 1.8% at 1 year, 14.9%

Table 3. Radiation therapy characteristics

Notched plaque	113 (79)
Circular plaque	30 (21)
Plaque size, mm	18.28 (2.22), 17.50
Scleral radiation, Gy	370 (115), 321
Radiation dose to optic nerve, Gy	72 (19), 64

Values are *n* (%) or mean (standard deviation), median. Subset analysis of 143 patients. Gy, Gray units.

by 5 years, and 32.8% by 10 years (Fig. 1d). The median time to death was 16.7 years. Of the patients who died, 17 (32.7%) were attributed to disease-specific mortality. Four patients died from a second malignancy. Cause of death was not available from the SSDI; however, for some patients, it was directly noted in the medical record. Event-free (metastasis or death) survival was 97.8%, 80.9%, and 61.7% at 1, 5, and 10 years, respectively (Fig. 1e). The median event-free survival was 9.2 years. A subset analysis of 143 patients with radiation data was performed (Table 3). Median plaque size was 17.5 mm, with most plaques being notched (79%). Median scleral radiation dose was 321 Gy, and median optic nerve dose for patients who received notched plaques was 64 Gy.

Table 4 describes the change in visual acuity over time. Median baseline BCVA was LogMAR of 0.1 (Snellen equivalent 20/25). At 1-year, median BCVA had declined to a LogMAR of 0.4 (Snellen equivalent 20/40). By the last follow-up exam, median BCVA was LogMAR 2.0 (Snellen equivalent count fingers).

Complications

Seven patients developed extraocular extension, all of whom ultimately died during the study period, four from metastases. More than half of the patients developed radiation retinopathy (*n* = 130) or optic neuropathy (*n* = 109). The median time to developing radiation retinopathy was 742 days (2.03 years) and optic neuropathy was 1,011.5 days (2.77 years).

Due to the regional referral nature of the practice, there was limited information available on treatments provided for either radiation retinopathy or optic neuropathy. Other observed complications included neovascularization of the iris (*n* = 20), vitreous hemorrhage (*n* = 62), serous retinal detachment (*n* = 84), and cataracts (*n* = 77). Fifty patients underwent cataract extraction.

Overall Survival Analysis

When simple variate Cox regression analysis was performed, there was no difference in survival due to gender, family, or personal history of other cancers, local recurrence, or laterality of the tumor. Race (white vs. non-white) was not evaluated as there were only six non-white patients. Age at surgery and initial tumor grade (AJCC 8 tumor, node, metastasis staging system, TNM) was significantly associated with survival (*p* < 0.001 for both). Smoking was significantly associated with overall survival (*p* = 0.042) (Table 5). But this association was nonsignificant after multivariate regression with adjustment to age at surgery (*p* = 0.13) and/or tumor stage (*p* = 0.067) (Table 6).

Risk Factors for Local Recurrence

For local recurrence, in the initial simple variate Cox regression model, there was no difference based on tumor height, maximum basal diameter, tumor grade, gender, race, or prior personal or family history of cancer. Two multivariate Cox models of time to local recurrence were used to evaluate risk factors for early recurrence, one with full model including tumor TNM stage but not the maximum basal diameter, the other with maximum basal diameter but not tumor stage. The analysis indicated that personal cancer history has a borderline significant association with earlier recurrence (*p* = 0.057 or 0.054 in each model, respectively, Table 7). Family history showed unexpected protective effect against early recurrence (*p* = 0.004 and 0.003 in each model, Table 7).

Risk Factors for Event-Free Survival

As shown in Table 8, sex and tumor stage are significantly associated with event-free (death or metastasis) survival. Females were significantly less prone to have events (HR = 0.06, *p* = 0.002 in the adjusted model). Additionally, tumor stage was also significantly associated with event-free survival, advanced stages exhibited much higher hazard for events to occur (HR = 3.71, *p* = 0.007). No other variables were associated with event-free survival.

Discussion

Juxtapapillary tumors present the treating ocular oncologist with challenges not otherwise encountered in the management of choroidal melanoma. For this reason, juxtapapillary melanomas were largely excluded from the landmark clinical trials which established plaque

Table 4. Visual acuity

Best Corrected Visual Acuity (BCVA) LogMAR, mean (standard deviation), median	
Baseline	0.47 (0.74), 0.10
1 Year	0.88 (0.92), 0.40
Final	1.56 (1.02), 2.00
Baseline BCVA LogMAR category, <i>n</i> (%)	
1	140 (61)
2	42 (18)
3	15 (6.5)
4	33 (14)
1-year BCVA LogMAR category, <i>n</i> (%)	
1	82 (38)
2	44 (21)
3	28 (13)
4	59 (28)
Final BCVA LogMAR category, <i>n</i> (%)	
1	31 (13)
2	25 (11)
3	37 (16)
4	138 (60)
Visual acuity at baseline, 1-year posttreatment, and final visit. <i>N</i> = 237.	

brachytherapy as the standard of care for most other forms of uveal melanoma. Since the publication of the COMS, only a few centers have retrospectively studied the efficacy of plaque brachytherapy specifically for this subset of tumors [3, 11–13]. These studies incorporated multiple different radiation sources (I-125, Ru-106, cobalt-60 [Co-60], and iridium-192 [Ir-192]) and plaque types (customized plaques vs. COMS) [11, 13] as well as adjuvant transpupillary thermotherapy [11]. Such differences in treatment modalities can make long-term outcomes and complication rates difficult to compare. Adding to this difficulty is the fact that the majority of juxtapapillary tumors are combined into larger overall uveal melanoma studies without specific subset analysis. The increased difficulty in treating juxtapapillary melanomas with appropriate margins due to their proximity to the optic nerve warrants consideration of these tumors as a potential subset of uveal melanoma with high risk of poor outcomes and complications. Our findings presented herein emphasize the challenges and limitations of treating juxtapapillary choroidal melanoma with brachytherapy.

A significant finding of this study – which corroborates similar observations from other centers – is that local control of juxtapapillary tumors following a single brachytherapy treatment is substantially decreased compared with other tumors. The COMS study

reported a local recurrence rate at 5 years of 10.6% and this is the most common benchmark to which other treatment center outcomes are compared. Since publication of the COMS study, however, multiple single institutions have reported 5-year recurrence rates of 5% or less [14–16]. Within this cohort of 237 juxtapapillary melanomas, the incidence of local recurrence at 5 years was comparatively greater at 10% and increased to 24.6% at 10 years. Shields et al. observed a similar trend in their cohort of 650 juxtapapillary tumors where local recurrence was diagnosed in 14% at 5 years and 21% at 10 years [11]. In a separate cohort of non-juxtapapillary tumors at the same center, local recurrence was only diagnosed in 3% at 5 years [17]. A Swedish study by Fili et al. [13] observed the same trend with local recurrence in 23% of juxtapapillary melanomas treated with brachytherapy. Hegde et al. [3] observed a 4-year local recurrence rate of only 7.6% in 34 patients with juxtapapillary or circumpapillary choroidal melanoma; however, a separate case series from this same center reported a 0% treatment failure rate in a separate cohort of 170 patients with choroidal melanoma undergoing fine-needle aspiration biopsy (who were presumably not limited to tumors adjacent to the optic nerve) [18].

Tumor proximity to the optic nerve is well established as a risk factor for local recurrence following brachytherapy [19]. Possible explanations include misalignment

Table 5. Simple regression for overall survival analysis

	N	HR	p value	95% CI
Sex	236	0.82	0.475	0.47–1.42
Age at surgery	236	1.06	0.000	1.04–1.08
Tumor stage	234	2.37	0.000	1.74–3.24
Personal cancer history	233	1.63	0.132	0.86–3.08
Family cancer history	230	0.89	0.691	0.48–1.62
Laterality	236	1.00	0.992	0.57–1.73
Local recurrence	234	1.33	0.421	0.66–2.67
Smoking	225	0.29	0.036¹	0.09–0.92
Race ²	220	NE	NE	NE

NE, not evaluated. ¹The apparent association of smoking with survival maybe due to an age effect as when smoking effect is adjusted to age at surgery, the regression *p* value is 0.13, and when adjusted to both tumor stage and age at surgery, the *p* value for smoking is 0.07. Also out of 44 smokers, there were only three death events; simple variate regression shows smoking appears to be associated with a better survival, but in multivariate regression with adjustment to age and tumor stage, the significance is lost (see Table 6). ²Estimate on race is out of range because there were only six non-white patients.

Table 6. Multivariate regression for overall survival^{1,2}

Variables included	HR	p value	95% CI
Model 1			
Sex	0.53	0.051	0.28–1.00
Age at surgery	1.05	0.000	1.03–1.08
Tumor stage ¹	2.31	0.000	1.61–3.31
Personal cancer history	1.30	0.494	0.61–2.77
Family cancer history	1.17	0.67	0.57–2.39
Local recurrence	1.36	0.664	0.34–5.41
Laterality	1.19	0.601	0.62–2.27
Smoking	0.28	0.042	0.08–0.96
TVC (local recurrence) ²	0.95	0.648	0.77–1.18
Model 2			
Age at surgery	1.04	0.000	1.02–1.06
Tumor stage	2.26	0.000	1.61–3.17
Smoking	0.33	0.067	0.1–1.08
Model 3			
Age at surgery	1.05	0.000	1.03–1.08
Smoking	0.4	0.127	0.12–1.3

¹AJCC, American Joint Committee on Cancer, 8th edition. Tumor Stage are according to tumor grade, T1 include T1a and 1b, and so on, until T4. ²Local recurrence was used as a time-varying covariate (TVC) in the multivariate Cox regression analysis.

Table 7. Simple variate regression for local recurrence

Variables included	HR	p value	95% CI
Univariate regression			
Sex	0.85	0.631	0.43–1.68
Age at surgery	1.01	0.484	0.98–1.03
Tumor T stage	1.02	0.913	0.70–1.50
Personal cancer history	1.66	0.201	0.76–3.58
Family cancer history	0.44	0.025	0.21–0.90
Laterality	0.93	0.845	0.47–1.85
Smoking	0.15	0.059	0.02–1.08
Maximum basal diameter	1.05	0.313	0.96–1.14
Multivariate regression			
Model 1^a			
Sex	0.69	0.345	0.32–1.49
Age at surgery	0.99	0.282	0.96–1.01
Personal cancer history	2.46	0.057	0.97–6.20
Family cancer history	0.31	0.004	0.14–0.69
Laterality	1.00	0.992	0.46–2.17
Smoking	0.16	0.069	0.02–1.15
Maximum basal diameter	1.00	0.950	0.90–1.12
Model 2^b			
Sex	0.69	0.337	0.32–1.48
Age at surgery	0.98	0.283	0.96–1.01
Tumor T stage	0.77	0.292	0.48–1.25
Personal cancer history	2.48	0.054	0.98–6.28
Family cancer history	0.30	0.003	0.13–0.67
Laterality	1.02	0.951	0.47–2.22
Smoking	0.15	0.064	0.02–1.11

^aModel 1 is a bivariate regression including AJCC tumor T stage with each covariate. ^bModel 2 is a bivariate regression including maximum basal diameter with each covariate.

of the plaque notch with the optic nerve sheath and loss of apposition between the plaque and the globe due to posterior tilt. Using intraoperative ultrasound, Murray et al. observed that plaque position was initially misaligned in 30% of peripapillary or posteriorly located tumors [16]. Local recurrence rate at 5 years was decreased to 1.7% with ultrasound-guided plaque localization, suggesting that the increased propensity of juxtapapillary tumors for treatment failure can be circumvented using this strategy. Visual outcomes in this cohort were also comparatively worse than non-juxtapapillary tumors. Median acuity declined to counts fingers or worse by the most recent follow-up exam. In the COMS by comparison, median visual acuity in patients with non-juxtapapillary tumors declined to only 20/125 at 3 years [20]. Sagoo et al. [12] also observed poorer visual outcomes in their juxtapapillary cohort with median visual acuity declining to 20/200 or worse. Likewise in a study of 34 juxtapapillary

Table 8. Risk factors for event-free (death or metastasis) survival

	Simple variate regression			Multivariate regression		
	HR	<i>p</i> value	95% CI	HR	<i>p</i> value	95% CI
Sex	0.33	0.028	0.12–0.89	0.06	0.002	0.01–0.38
Age at surgery	1.00	0.863	0.97–1.03	0.98	0.234	0.94–1.01
Tumor T stage	2.18	0.014	1.17–4.07	3.71	0.007	1.42–9.68
Personal cancer history	1.35	0.567	0.48–3.81	1.95	0.378	0.44–8.61
Family cancer history	0.71	0.490	0.27–1.86	0.45	0.224	0.13–1.62
Laterality	1.12	0.796	0.47–2.69	0.72	0.607	0.20–2.53
Local recurrence	1.46	0.450	0.55–3.85	6.22	0.153	0.51–76.3
Smoking	2.47	0.237	0.55–11.06	1.15	0.882	0.19–6.91
TVC (local recurrence) ^a				0.68	0.263	0.35–1.33

^aLocal recurrence was used as a time-varying covariate (TVC) in the multivariate Cox regression analysis.

melanomas, Hegde et al. [3] observed 20/200 or better visual acuity in only 23.5% of patients at final follow-up, while 26.5% had declined to light perception or no light perception. Patients undergoing brachytherapy for juxtapapillary melanomas should be counseled that the risk of vision loss after treatment is greater than that for other tumors. Proton beam radiotherapy is an alternative to brachytherapy which may mitigate the limitations of brachytherapy in treating juxtapapillary melanoma. Lane et al. [21] report a large study of 573 patients treated with proton beam therapy that found a lower rate of local recurrence than our study with 3.3% at 5 years and 6% at 10 years (vs. 10% and 24.6% at 5 and 10 years). Unfortunately, proton beam therapy is limited to only a few centers and is not widely accessible. Moreover, proton beam therapy has a higher rate of radiation retinopathy and optic neuropathy compared to brachytherapy [21, 22]. Enucleation rates were also similar (11.5% vs. 12%) with the majority performed for neovascular glaucoma in those receiving proton beam therapy [21].

There are several limitations to this study, and specifically, this is a moderately sized retrospective study from a single institution. As this is a single referral center for seven states, many patients received follow-up closer to home for radiation-related complications, limiting the ability to report on the timing, treatment, and resolution of these complications. The median follow-up time of 5 years may miss later complications including recurrences and metastasis with associated metastasis-related death. Confounding

elements in our study include the natural progression of cataracts, specifically regarding our finding of 32.5% of patients in a predominantly older population. The majority of juxtapapillary tumors were primarily centered in the posterior pole ($n = 224/237$, 96%), limiting the discussion around specific implications and risk for death for anterior and ciliary body tumors ($n = 9/237$). The lack of reports with similar statistical studies and treatments makes direct comparisons difficult. Additional limitations to our study include the tumor height range, base diameter, and a lack of genetic information. There is likely wide genetic variation in these tumors, leading to differing patient outcomes. Yet, we do not report any biopsy-related tumor pathology, genetic testing (including chromosomal analysis, gene expression profile, or preferentially expressed antigen in melanoma positivity) due to the low number of biopsies performed in this cohort. This limits prognostic comparisons to more recent patient cohorts with genetic data available.

This retrospective single site cohort of 237 juxtapapillary uveal melanoma patients treated with I-125 plaque brachytherapy demonstrated higher rates of local recurrence and vision loss compared with other tumor configurations. Cumulative incidence of enucleation, distant metastasis, and overall mortality were otherwise comparable to COMS and other prior studies of I-125 and Ru-106 brachytherapy. Juxtapapillary uveal melanomas remain challenging to treat regardless of the chosen modality, and individualized counseling based on patient tumor characteristics is recommended.

Statement of Ethics

This study was reviewed and approved by the Institutional Review Board at the University of Tennessee Health Science Center, Approval No. 16-04640. Written informed consent was waived by the Institutional Review Board at the University of Tennessee Health Science Center.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Conception and design: Ilyse Kornblau, Andrew Wiles, Enrique Izaguirre, Fang Liu-Smith, and Matthew Wilson. Analysis and interpretation: Ilyse Kornblau, Benjamin King, Fang Liu-Smith, and Matthew Wilson. Data collection: Ilyse Kornblau, Nikolas Hopkins, Benjamin King, and Matthew Wilson. Obtained funding: not applicable.

Data Availability Statement

The data support the research are IRB approved and HIPAA protected to the institution from which they were derived. Deidentified portions of the data are available upon request from the corresponding author (Matthew W. Wilson, MD, University of Tennessee Health Science Center, Department of Ophthalmology, 930 Madison Ave, Suite 100, Memphis, TN 38104. Email: mwilson5@uthsc.edu) and approval from the University of Tennessee Health Science Center.

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